Mid-Semester Report
Term Project

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Title

Material Balances for Ammonia Production via Haber Process with Recycle and Purge

Abstract

The objective of this project is to perform detailed material balance calculations for the ammonia synthesis process using the Haber process, which includes recycle and purge streams. The process is inherently limited by low single-pass conversion, making the inclusion of recycle essential to increase overall yield. A purge stream is required to avoid inert gas accumulation. This report presents the process flow diagram, governing material balance equations, and the effect of reactor conversion and purge ratio on ammonia yield.

Introduction

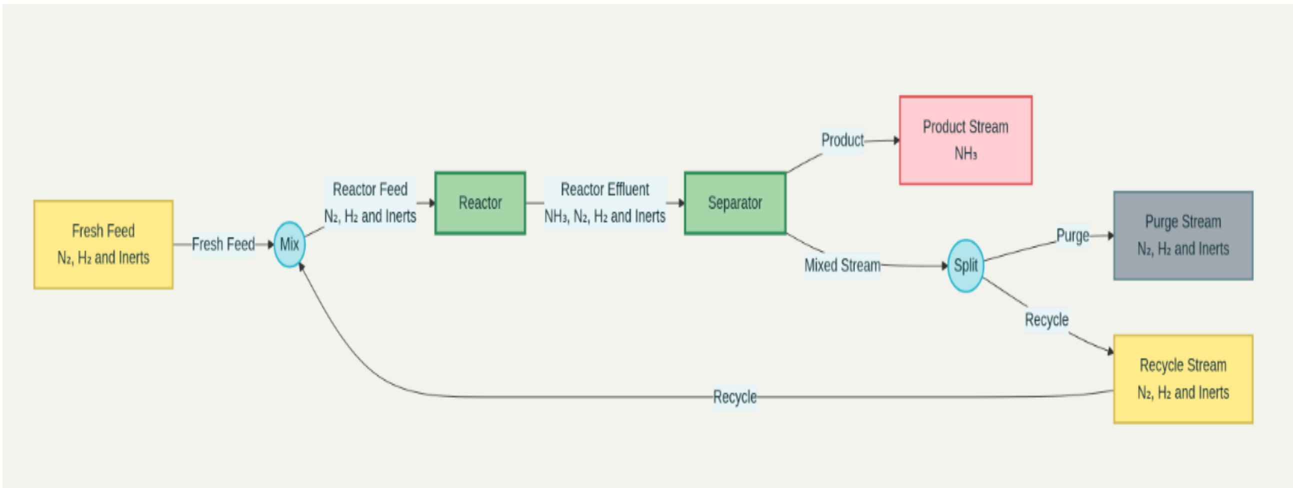
Ammonia is an essential industrial chemical, widely used in fertilizers and chemical synthesis. Its production is dominated by the Haber process, where nitrogen reacts with hydrogen at high pressure and temperature in the presence of a catalyst. Due to equilibrium limitations, only 20–30% of the feed reacts in a single pass. Therefore, unreacted gases are separated, recycled, and partially purged to avoid buildup of inerts. This project focuses on material balance derivations for such a system.

Process Flow Diagram

The process consists of the following units:
1. Fresh Feed: Nitrogen, hydrogen, and small fraction of inerts.
2. Reactor: Converts a fraction of N_2 and H_2 into NH_3.
3. Separator: Ammonia is removed; unreacted gases exit.

4. Recycle Stream: A fraction of gases are sent back to the reactor.
5. Purge Stream: A small fraction is purged to prevent inert accumulation.

A detailed schematic for the process is given below:



Material Balance Derivations

Assumptions:

Basis: 100 mol of fresh feed (F).

Fresh Feed Composition:

Fresh N2 = F_{N2} mol, Fresh H2 = F_{H2} mol , Inerts = F_I mol

Constraint: $F_{N2} + F_{H2} + F_I = 100$.)

- **Reaction :** $N_2 + 3H_2 \text{->} 2NH_3$
- **Single-Pass Conversion (x):** Fraction of N_2 entering the reactor that is consumed. (Base Case: x=0.25)
- **Purge Ratio (p):** Fraction of the separator overhead stream that is purged. (Base Case: p=0.10)
- **Separation:** Ideal; all produced NH_3 is removed as a liquid product. Unreacted N_2, H_2, and inerts are not dissolved in the product.
- **Steady State:** Accumulation is zero for all units.

System Setup:

The overall system is solved using the steady-state balance around the entire process, which eliminates the recycle stream. The key is recognizing that the purge stream composition is identical to the composition of the stream leaving the separator and entering the splitter.

Overall Balances:

At steady state, for the entire system (excluding the recycle):

- Input = Output

The balances for each component are:

1. **Inert Balance (I):**
   $F_I = P_I$ ...(1)
   (All inerts entering with the fresh feed must leave via the purge.)
2. **Nitrogen Balance (N_2):**
   $F_{N2} = N_{2,reacted} + P_{N2}$ ...(2)
3. **Hydrogen Balance (H_2):**
   $F_{H2} = 3 \times N_{2,reacted} + P_{H2}$ ...(3)
   (From reaction stoichiometry, H_2 consumed is 3 times N_2 consumed.)
4. **Ammonia Balance (NH_3):**
   $N_{NH3,product} = 2 \times N_{2,reacted}$ ...(4)

Linking Purge and Reactor Composition:

Let S be the total molar flow rate of the stream leaving the separator (and entering the splitter). This stream contains only N_2, H_2, and I. The mole fraction of any component j in stream S is $y_j = n_{j,S} / S$.

The purge stream P is a fraction p of S, so for any component:

$$P_j = p \times n_{j,S} .$$

However, at steady state, the flow rate of any inert component leaving the separator must equal its flow rate entering the separator. The inerts enter the separator only from the reactor outlet, and they are not reacted. Therefore, the flow of inerts in stream S is constant throughout the recycle loop and is equal to the fresh feed input F_I.

$$P_I = p \times F_I$$

But from equation (1), $F_I = P_I$. Substituting:

$$F_I = p \times F_I$$

This equation only holds if $p = 1$, which is incorrect. The flaw is that the flow of inerts in stream S is **not** F_I, but rather F_I/p. This is because the inerts accumulate in the recycle loop until their rate of entering (via fresh feed) equals their rate of leaving (via purge).

$$\text{Inerts in Stream S} = F_I/p$$

The flow of N_2 in stream S is the N_2 that was not converted in the reactor.
The N_2 entering the reactor is $[F_{N2} - (1-p)P_{N2}]/p$ (from an overall balance around the mixer and reactor). Solving this simultaneously with other equations gives the final flow rates.

Solution Approach:

The system of equations (1) to (5) and the relationship between reactor inlet and fresh feed is solved simultaneously. For a given F_{N2}, F_{H2}, F_I, x, p one can calculate:

- $N_{2,reacted}$ from the inert balance and reactor inlet.
- $N_{NH3,product}$ from equation (4).
- All other flow rates.

Results and Discussion

The effect of reactor conversion (x) and purge ratio (p) on ammonia yield was studied by solving the system of material balance equations detailed above. For a fresh feed containing 24 mol N_2, 72 mol H_2, and 4 mol inerts (to satisfy the 3:1 H_2:N_2 stoichiometry), the results are as follows:

Higher single-pass conversion directly increases ammonia production per pass, reducing the required recycle flow and thus minimizing losses in the purge. However, due to thermodynamic equilibrium limitations, this conversion cannot be increased arbitrarily beyond 25-30% under practical conditions.

The purge stream is essential to prevent the accumulation of inerts, which would dilute the reactor feed, increase compression costs, and lower the reaction rate.

However, a higher purge ratio (p) leads to greater losses of unreacted hydrogen and nitrogen, thereby decreasing the overall process yield.

For example, solving the material balances for the base case:

- At x = 25% and p = 10% , the process yields **14.8 mol of NH_3** per 100 mol of fresh feed.
- At x = 30% and p = 5%, the yield increases to approximately **21.5 mol of NH_3.**
- At x = 20% and p = 15%, the yield decreases to approximately **10.5 mol of NH_3.**

These results highlight the critical operational trade-off: while higher conversion and lower purge ratios independently improve yield, a minimum purge is necessary to avoid inert build-up. The yields obtained are significantly lower than the theoretical maximum without inerts (~38.5 mol NH_3), underscoring the substantial impact of inert management and purge losses on process economics.

Conclusion

This project developed a steady-state material balance model for an industrial ammonia synthesis loop, incorporating key features such as reactor conversion, product separation, and recycle with purge. The analysis underscores that the recycle stream is crucial for maximizing overall yield, while the purge stream is necessary to control inert concentrations. The results clearly quantify the operational trade-off between ammonia yield and inert management, showing that yield is positively dependent on single-pass conversion and inversely related to the purge ratio.

The framework established here successfully calculates flow rates for all key process streams and provides critical insights into the process's constraints. For the final report, this model will be extended to include a comprehensive sensitivity analysis, graphical representations of the trade-offs, and a comparison with typical industrial data to validate the theoretical findings.

References

1. R. M. Felder and R. W. Rousseau, Elementary Principles of Chemical Processes, Wiley.

2. Himmelblau, D. M., Basic Principles and Calculations in Chemical Engineering.
3. J. M. Smith, H. C. Van Ness, and M. M. Abbott, Introduction to Chemical Engineering Thermodynamics.